

Ecological Succession

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

Cue focus	Ecological succession is the gradual change in community composition over time; primary succession can begin on bare rock, while later stages often support greater soil development, plant structure, and biodiversity.
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Round 1 - Retrieve It

1. What is ecological succession?

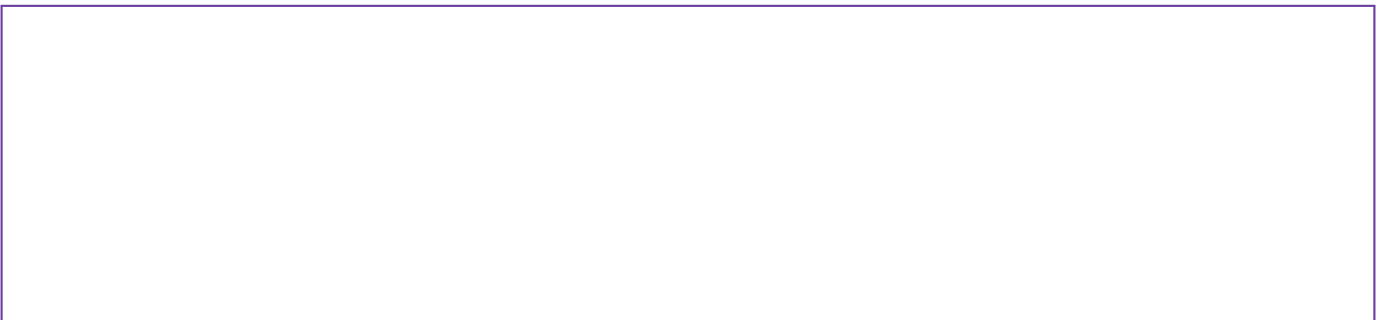
2. What kind of surface can primary succession begin on?

3. What are pioneer species?

4. What general biodiversity trend often occurs as succession proceeds?

Round 2 - Sketch It

Draw four stages from bare rock to a mature community. Include pioneer species, grasses/small plants, shrubs, and larger plants/trees where appropriate.



Ecological Succession - Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 - Apply the Relationship

1. New lava cools into bare rock with no soil. Primary or secondary succession?

2. A wildfire removes vegetation but leaves soil. Primary or secondary succession?

3. Why are lichens often used as an example of pioneer species on bare rock?

Round 4 - Reason from Evidence

Compare what makes primary succession usually slower than secondary succession.

Ecological Succession - ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A

Articulate

Explain Ecological Succession in your own words. Use one example that is not printed on the cue card.

C

Connect

Connect succession to disturbance. How does the amount of soil and surviving life after a disturbance affect recovery?

E

Extend

A construction site is abandoned after topsoil remains but most plants are removed. Predict the likely succession type and early changes.

Confidence check Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it

TEACHER KEY - Ecological Succession

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. A gradual change in the species/community structure of an ecosystem over time.
2. Bare rock or another surface with little/no soil.
3. Early colonizers that can live in harsh conditions and help change the environment for later species.
4. Biodiversity and structural complexity often increase, though real ecosystems can vary.

Sketch It - look for

A reasonable progression showing soil/community development and increasing complexity. Exact species depend on ecosystem.

Apply It

1. Primary succession.
2. Secondary succession.
3. They can colonize harsh surfaces and help weather rock/build organic material, supporting soil development.

Reason from Evidence - look for

Primary starts without developed soil/seed bank and must build conditions from bare substrate. Secondary begins with soil and often surviving organisms/seeds, speeding recovery.

ACE - Extend look-for

Secondary succession; soil remains, so grasses/weeds and other early colonizers can establish relatively quickly, followed by later community changes.

Suggested use

Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.